

Statistical Report

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Statistical Report on Fertility Data in Singapore

Abstract

Introduction

Can time series models trained on data from 1960–2012 accurately forecast Singapore’s continuing fertility decline over 2013–2024?

Methods

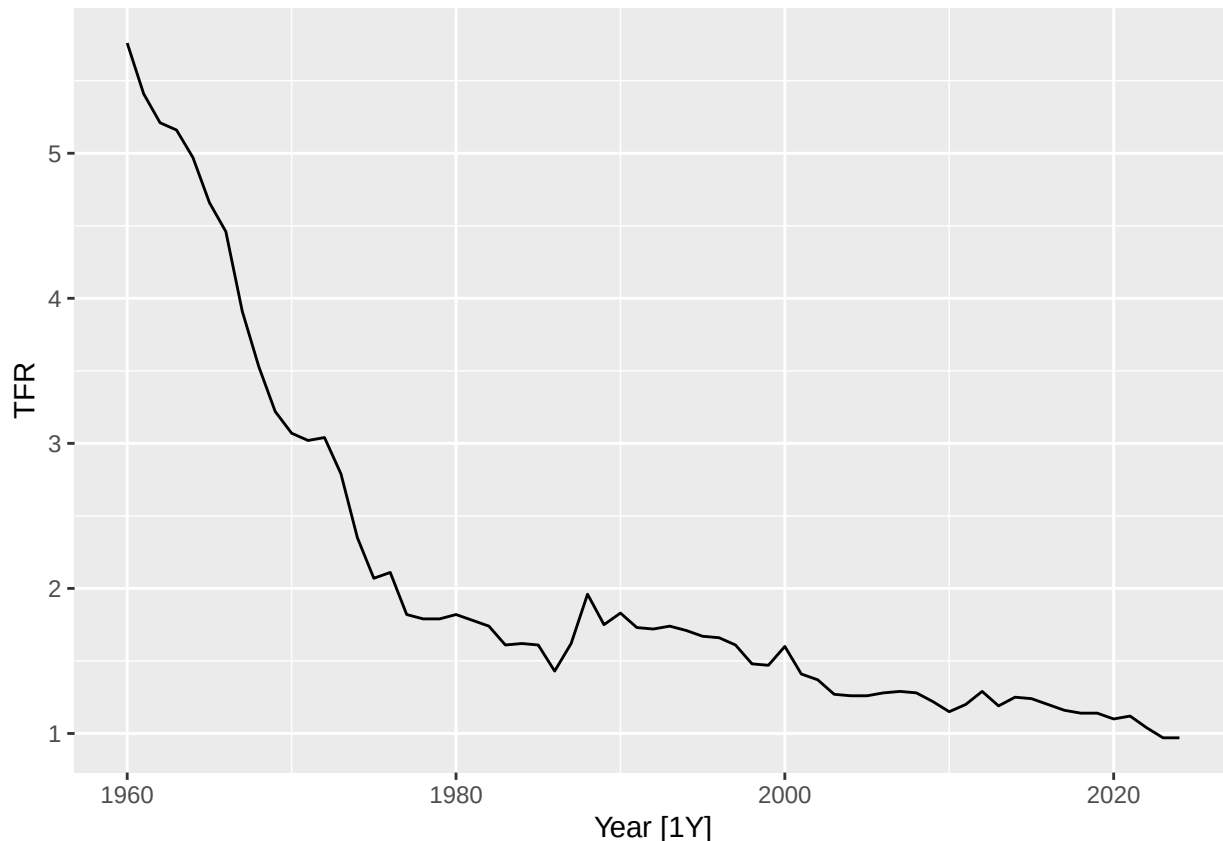
3.1 Data preparation

- Describe TFR and TLB.
- Explain reshaping into annual time series.
- Define training period 1960–2012 and testing period 2013–2024.

The modelling workflow was designed to answer our forecasting needs. The TFR and TLB series were extracted from the dataset, reshaped into annual time series from 1960–2024, and split into a training period from 1960–2012 and a testing period from 2013–2024.

3.2 Transformation and stationarity assessment

- Explain why stationarity matters for ARIMA modelling.
- Mention original, log, differenced, second-differenced, and log-differenced series.
- Explain KPSS test.
- Justify log transformation for variance stabilisation.
- Justify differencing for trend removal.
- State that log-first-differencing was retained as a primary candidate transformation, while second-differenced models were considered as comparisons.



Both TLB and TFR were then inspected running a series of tests to ensure stationarity and appropriateness of the series for ARIMA modeling.

3.3 Model identification

- Explain use of ACF and PACF.
- State that transformed series showed dependence around lags 11–13.
- Explain why low-order ARIMA models were used only as baselines.
- Explain why higher-lag ARIMA and SARIMA period-12 models were considered.

3.4 Model estimation and diagnostics

- Explain models fitted on training data only.
- State model viability criteria: white noise residuals, residual ACF/PACF, Ljung–Box.
- State that models with remaining lag-12 residual autocorrelation were rejected.

3.5 Forecast evaluation

- Forecast 2013–2024.
- Compare with observed values.
- Use RMSE, MAE, SSE/AICc.
- Select final model based on residual viability first, then forecast accuracy and simplicity.

- What the data is
-
- talk about how the model's were chosen
 - Looked at transformation to make the data stationary
 - Then chose to log diff as it had the best __ stationarity i think
 - * The log transformation addresses **variance non-stationarity**, and the differencing addresses **mean non-stationarity** (the trend). They solve different problems. The KPSS test would confirm mean-stationarity after differencing, not “best stationarity” between log-diff

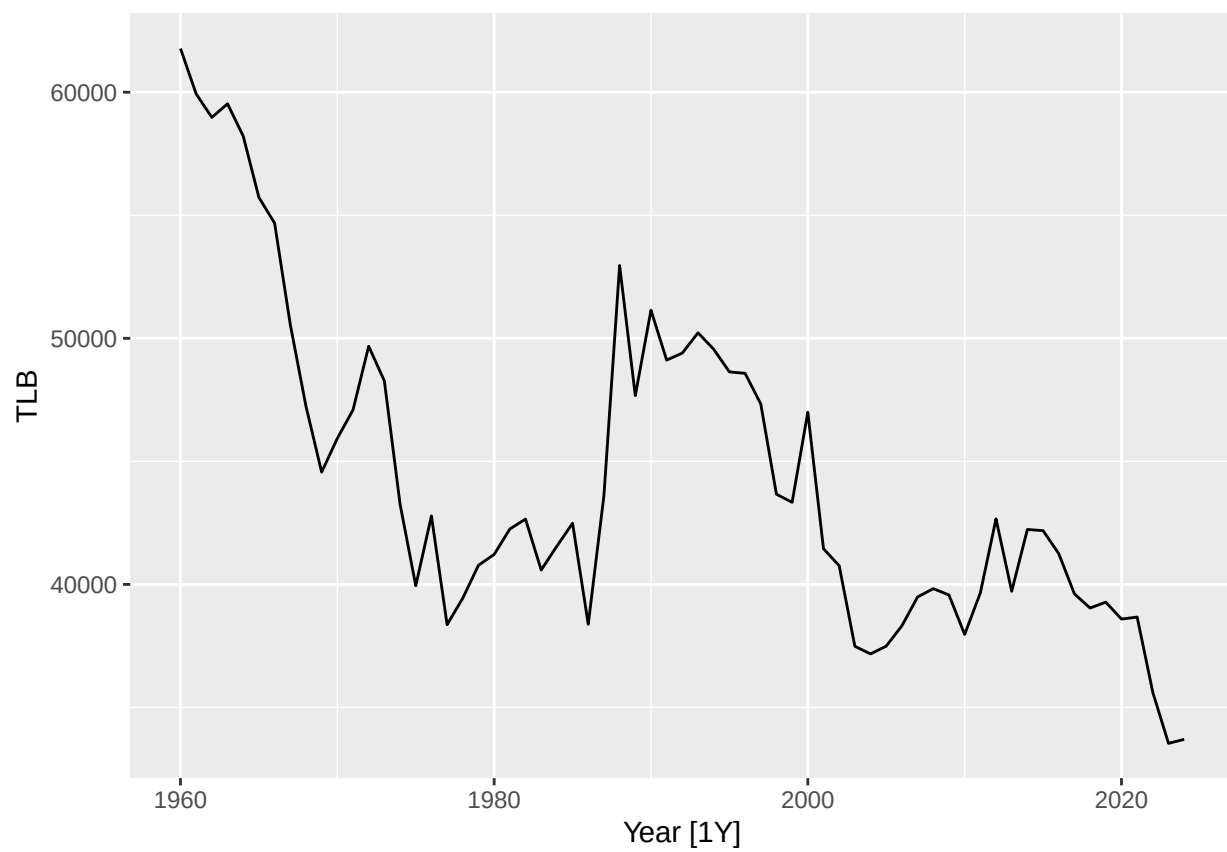


Figure 1: Total Live Births in Singapore, 1960–2024

and plain-diff.

* **Is the log transform justified?**

Yes, clearly. TFR’s coefficient of variation drops from **0.47 to 0.16** between the early and late periods — the variance shrinks dramatically as the level drops. You can see this in the first-diff plot (large swings in 1960–1975, small swings after). The log-diff plot has much more stable amplitude across the full series. For TLB, the case is weaker (CV goes from 0.16 to 0.11) but the log doesn’t hurt and keeps both series on the same scale.

Is one difference enough?

This is the one wrinkle. The KPSS results:

Transformation	TFR p-value	TLB p-value
First diff	0.01 (reject)	0.10 (pass)
Log + first diff	0.04 (borderline)	0.10 (pass)
Double diff	0.10 (pass)	0.10 (pass)

For TFR, log-diff is borderline at $p = 0.04$ — technically still rejects KPSS at the 0.05 level. Double differencing passes cleanly. This is why `ARIMA()` auto-selected $d = 2$ for TFR.

However, your SARIMA models with $d = 1$ on the log scale **pass the Ljung-Box test** and produce good forecasts. The KPSS test with only 53 annual observations has limited precision, and the borderline $p = 0.04$ isn’t damning. Double differencing (bottom plot) looks noisier and risks over-differencing.

Bottom line

Log + first difference is defensible — the log is well-justified by variance stabilisation, and $d = 1$ is adequate given that the fitted models achieve white noise residuals. In your report, you should acknowledge the borderline KPSS result for TFR and explain that you chose $d = 1$ because (a) the fitted SARIMA models pass Ljung-Box, and (b) double differencing risks over-differencing with a short series.

- The data still had trend for $yr=12$ (zodiac i think)
- So informed decision to try ARIMA (12,x, y) and SARIMA models potentially

Series	Models
TFR	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(1,2,1)(1,0,0)[12]
TLB	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(0,2,1)(1,0,0)[12]

Results

Discussion

Conclusion

References

Appendices if applicable.